



SUBHAJIT PAL

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ABOUT

Full-time **Ph.D.** scholar specializing in **Quantum Communication** and **Post-quantum Cryptography**. Research focuses on quantum-safe network design and the security of QKD protocols. **Two years** of experience in academic research and applied quantum cryptography projects. Cleared **UGC NET-JRF** and holds industry certifications in PQC. Passionate about bridging theoretical foundations with real-world secure communication.

EDUCATION

Ph.D. (Science) in Information Technology | University of Calcutta 2024 – 2029 | Registered- Dec. 2025

Tentative Thesis Title: *Methodologies for the Quantum-Safe Networks Using Quantum Communication Protocols*

Research Objectives: (1) Develop a Hybrid Quantum-Safe Communication Framework (2) Analyse Security Vulnerabilities and Their Impact on Hybrid Systems (3) Develop Mitigation Strategies for Identified Vulnerabilities

M.Sc. in Computer Science Visva-Bharati University 83.5% 2017 – 2019

B.Sc. in Computer Science The University of Burdwan 78% 2014 – 2017

B.Ed. The University of Burdwan 84.8% 2019 – 2021

Higher Secondary (12th) — Science WBCHSE 68.6% 2012 – 2014

Secondary (10th) WBBSE 79.42% 2011 – 2012

PROFESSIONAL EXPERIENCE

Research & Development Intern Sep. 2025 – Feb. 2026 (6 months)

QClairvoyance Quantum Labs Pvt. Ltd., Hyderabad (Remote)

- Contributed to the **Quantum Private CA Project** — post-quantum cryptography and PKI infrastructure for quantum-secure certificate authorities.
- Developed certificate revocation logic and anomaly detection in quantum-secure PKI systems.

PUBLICATION

[1] Subhajit Pal, Ajanta Das & Amlan Chakrabarti, “**Q-email: A Conference Key-based Quantum Safe Encryption Technique for End-to-End Group Communication**,” *QuantumKol-2025 (Intl. Conference on Quantum Technology and Applications)*, Kolkata, 17–19 Dec. 2025. [Accepted]

ACADEMIC CREDENTIALS & PROFESSIONAL CERTIFICATIONS

- UGC NET-JRF — CS & Applications (2022)
- NPTEL Blockchain & Applications — Top 5% (2022)
- WB TET (2023) & CTET Paper I & II (2023)
- Amazon Braket Quantum App Dev — AWS (2024)
- WB SET — CS & Applications (2020 & 2023)

TECHNICAL SKILLS

Programming Languages	C, C++, Python, SQL, HTML, JavaScript
Frameworks & Tools	Amazon Braket, Qiskit, Ibm Quantum, CDAC Qniverse
Core Computer Science	Data Structures & Algorithms, Theory of Computation, DBMS, Operating System, Computer Architecture, Cryptography & Network Security, Software Engineering, Data Communication, Compiler Design, Graph Theory
Specialized & Emerging Tech	Quantum Computing, Quantum Communication Protocols, Post-Quantum Cryptography, Blockchain, Machine Learning, Data Mining, AI